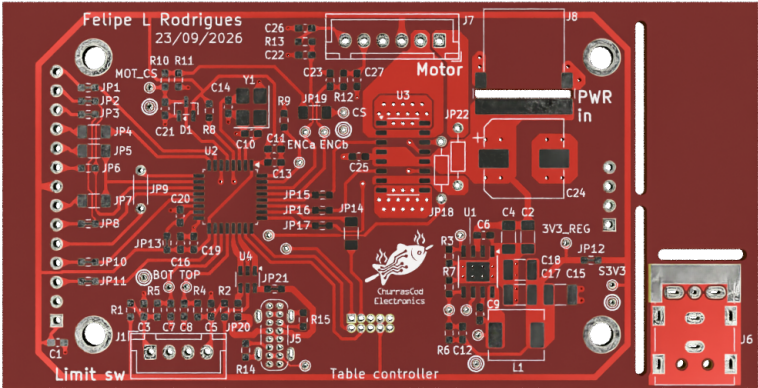
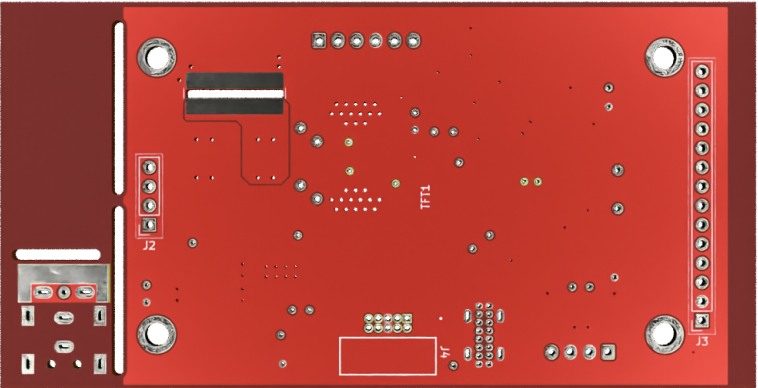


Desk controller fabrication drawing

Top view



Bottom view



	Desk controller		Sheet: Bare board overview	
	Notes		File: ktest.kicad_pcb	
	c1		Rev: 0.0.2	Drwn: FLR
	c2		Mat: Concept	Rev'd: FLR
	c3		Date: 06/10/2026	RI's'd: FLR
c4		Generated with KiCad	Size: A4	Page: 1/14

Fabrication requirements

Fabrication requirements

IPC

In case of divergences, dimensions from Gerber files take precedence

Surface finish: Lead-free HASL

Copper thickness: minimum x um external layer, minimum x internal layer, per IPC xxx

vias to be left untented / tented / filled and capped

process must be RoHS compliant

flammability rating shall meet as a minimum ul94v-0

tg170 or better

tolerance xxx or as per ipc

Board build up

	Material	Layer	Thickness	Dielectric	Type
		F.Paste			Paste Mask
		F.Silkscreen			Legend
		F.Mask	0.01mm		Solder Mask
	Copper	F.Cu	0.035mm (1.00oz)		Signal
	Prepreg		0.1mm	FR4	Dielectric
	Copper	In1.Cu	0.035mm (1.00oz)		Signal
	Core		1.24mm	FR4	Dielectric
	Copper	In2.Cu	0.035mm (1.00oz)		Signal
	Prepreg		0.1mm	FR4	Dielectric
	Copper	B.Cu	0.035mm (1.00oz)		Signal
		B.Mask	0.01mm		Solder Mask
		B.Silkscreen			Legend
		B.Paste			Paste Mask

Total thickness: 1.6mm



Desk controller

Notes

c1
c2
c3
c4

Sheet: Fabrication requirements

File: ktest.kicad_pcb

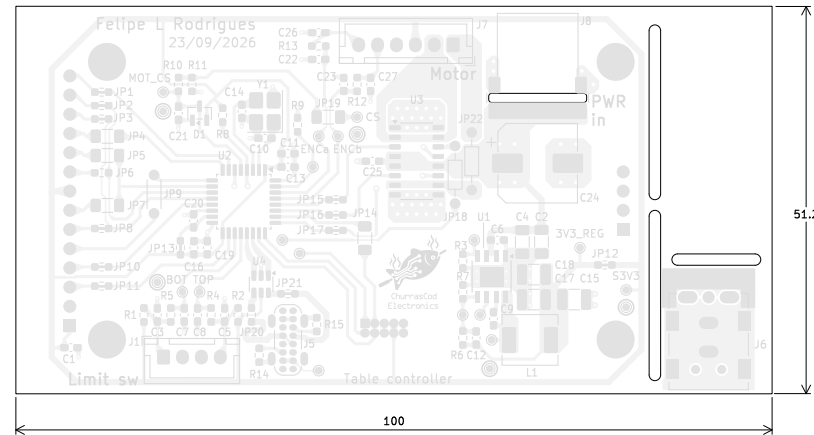
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Mat: Concept
Date: 06/10/2026

Dwg: NA
Generated with KiCad

Drwn: FLR
Rev'd: FLR
Rls'd: FLR

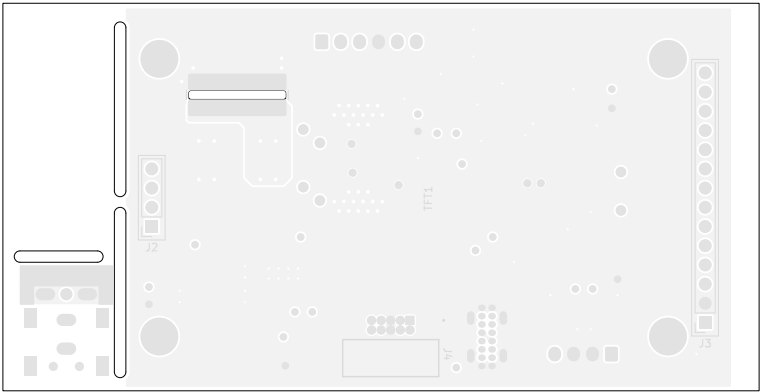
Size: A4
Page: 2/14

Top fabrication (scale 1:1)



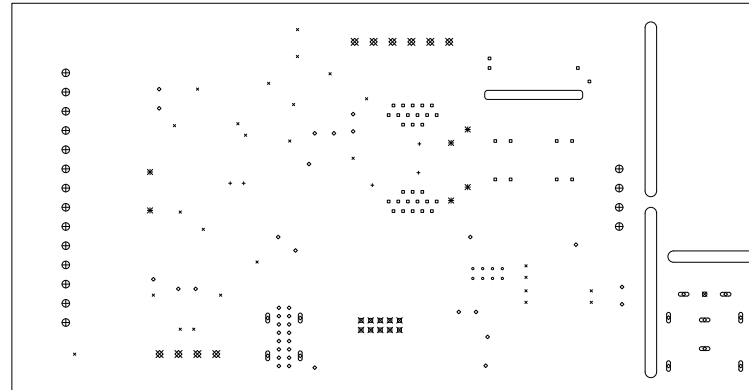
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	c2		Mat: Concept	Rev'd: FLR
	c3		Date: 06/10/2026	RLs'd: FLR
c4		Dwg: NA	Size: A4	
Generated with KiCad			Page: 3/14	

Bottom fabrication (scale 1:1)



Desk controller		Sheet: Bottom fabrication (scale 1:1)	
		File: ktest.kicad_pcb	
Notes		Rev: 0.0.2	Drwn: FLR
		Mat: Concept	Rev'd: FLR
		Date: 06/10/2026	RLs'd: FLR
		Dwg: NA	Size: A4
		Generated with KiCad	Page: 4/14

Drill drawing — L1 - L4 (scale 1:1)



Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair
×	26	0.30mm	PTH	Round	F.Cu – B.Cu
○	8	0.30mm	PTH	Round	F.Cu – B.Cu
+	5	0.40mm	PTH	Round	F.Cu – B.Cu
□	40	0.50mm	PTH	Round	F.Cu – B.Cu
◇	37 + 12	0.50mm	PTH	Round + Slot	F.Cu – B.Cu
⊗	1	0.60mm	PTH	Round	F.Cu – B.Cu
✱	6	0.70mm	PTH	Round	F.Cu – B.Cu
⊗	10	0.74mm	PTH	Round	F.Cu – B.Cu
⊗	10	0.95mm	PTH	Round	F.Cu – B.Cu
⊕	18	1.00mm	PTH	Round	F.Cu – B.Cu
Total 173					



Desk controller

Notes

c1
c2
c3
c4

Sheet: Drill drawing — L1 - L4 (scale 1:1)

File: ktest.kicad_pcb

Rev: 0.0.2
Mat: Concept
Date: 06/10/2026

Dwg: NA
Generated with KiCad

Drwn: FLR
Rev'd: FLR
Rls'd: FLR
Size: A4
Page: 5/14

Drill drawing — L1 - L4 (scale 1:1)



Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair
×	2	1.20mm	NPTH	Round	F.Cu – B.Cu
○	4	3.20mm	NPTH	Round	F.Cu – B.Cu
	Total 6				



Desk controller

Notes

c1
c2
c3
c4

Sheet: Drill drawing — L1 - L4 (scale 1:1)

File: ktest.kicad_pcb

Rev: 0.0.2

Mat: Concept

Date: 06/10/2026

Dwg: NA

Generated with KiCad

Drwn: FLR

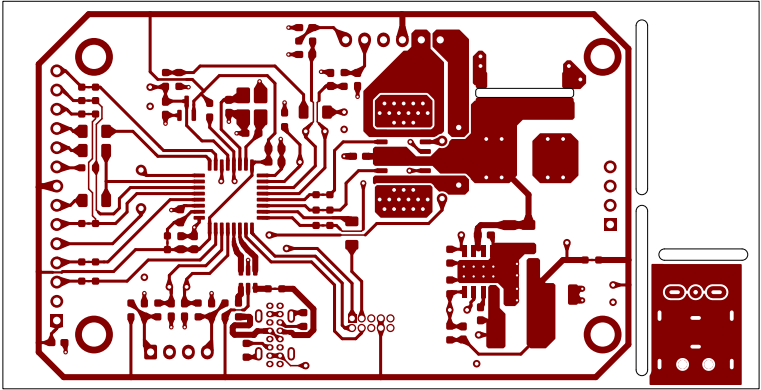
Rev'd: FLR

RLs'd: FLR

Size: A4

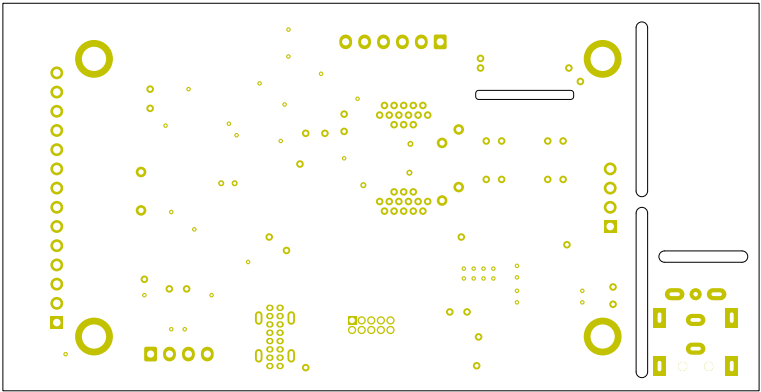
Page: 6/14

F.Cu (scale 1:1)



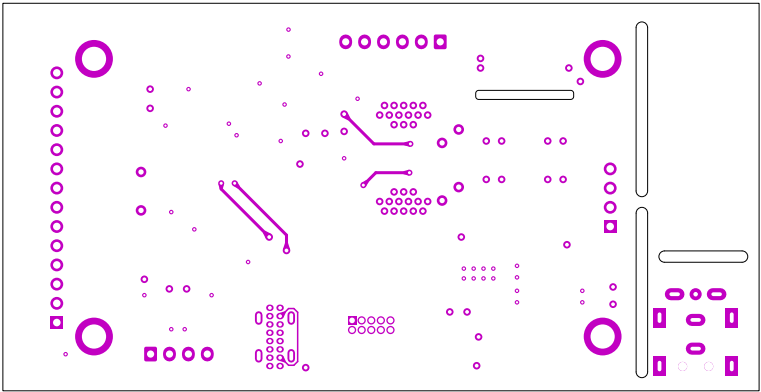
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	Notes c1 c2 c3 c4		File: ktest.kicad_pcb	
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			Mat: Concept	Rev'd: FLR
			Date: 06/10/2026	RLs'd: FLR
			Dwg: NA	Size: A4
			Generated with KiCad	Page: 7/14

In1.Cu (scale 1:1)



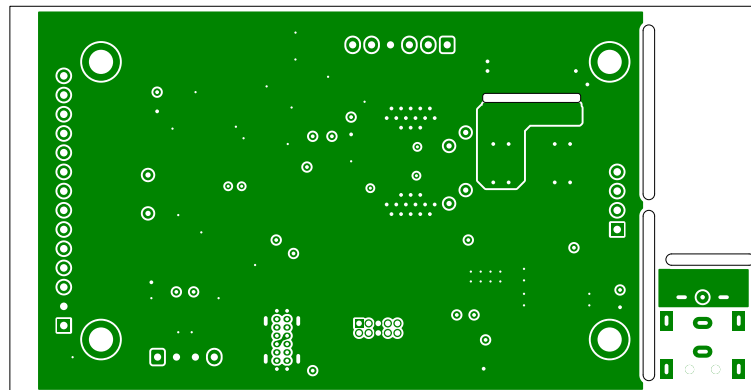
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		File: ktest.kicad_pcb	
Notes c1 c2 c3 c4	Rev: 0.0.2		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date: 06/10/2026		RLs'd: FLR
	Dwg: NA		Size: A4
		Generated with KiCad	Page: 8/14

In2.Cu (scale 1:1)



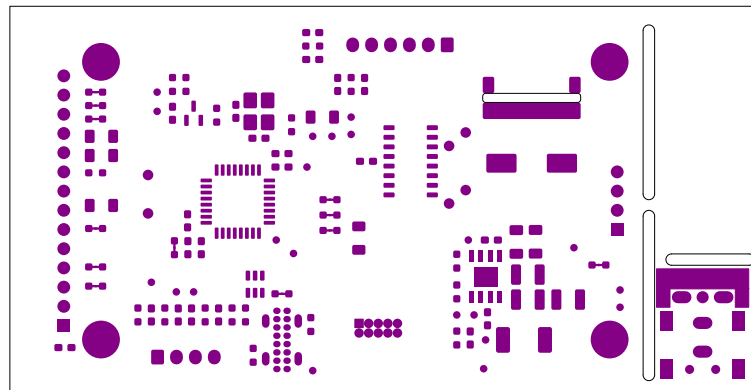
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	Mat: Concept	Rev'd: FLR
	Date: 06/10/2026	RLs'd: FLR
Notes	Dwg: NA	Size: A4
	Generated with KiCad	Page: 9/14

B.Cu (scale 1:1)



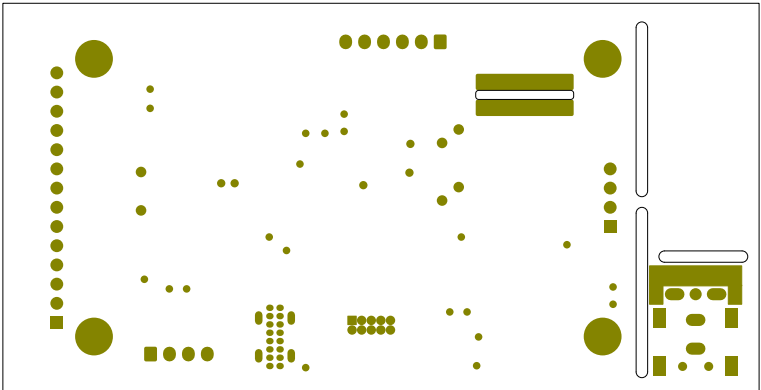
	Desk controller		Sheet: B.Cu (scale 1:1)	
	Notes c1 c2 c3 c4		File: ktest.kicad_pcb	Drwn: FLR
			Rev: 0.0.2	Rev'd: FLR
			Mat: Concept	RLs'd: FLR
			Date: 06/10/2026	Size: A4
			Dwg: NA	Page: 10/14
			Generated with KiCad	

F.Mask (scale 1:1)



	Desk controller		Sheet: F.Mask (scale 1:1)	
	Notes		File: ktest.kicad_pcb	
	c1		Rev: 0.0.2	Drwn: FLR
	c2		Mat: Concept	Rev'd: FLR
	c3		Date: 06/10/2026	RLs'd: FLR
c4		Dwg: NA	Size: A4	
Generated with KiCad			Page: 11/14	

B.Mask (scale 1:1)



Desk controller		Sheet: B.Mask (scale 1:1)	
		File: ktest.kicad_pcb	
Notes c1 c2 c3 c4	Rev: 0.0.2		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date: 06/10/2026		RLs'd: FLR
	Dwg: NA		Size: A4
		Generated with KiCad	Page: 12/14

Felipe L Rodrigues
23/09/2026

C26
R13
C22

J7 J8

R10 R11

MOT_CS Y1

C23 C27 Motor

JP1 JP2 JP3 JP4 JP5 JP6

C14 C21 D1 R8 U2 C10 C11 ENCa ENCb C13 C25

JP9 JP7 JP8 JP15 JP16 JP17 JP14

C20

JP13 C19

JP10 JP11

C16 B07 TOP R5 R4 R2

U4 JP21

R1 J1 C3 C7 C8 C5 JP20 J5 R14

C25 U3 JP22

PWR in

JP18 U1 C6 C4 C2 3V3_REG JP12 C18 C17 C15 S3V3

R3 R7

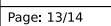
C9 R6 C12 L1

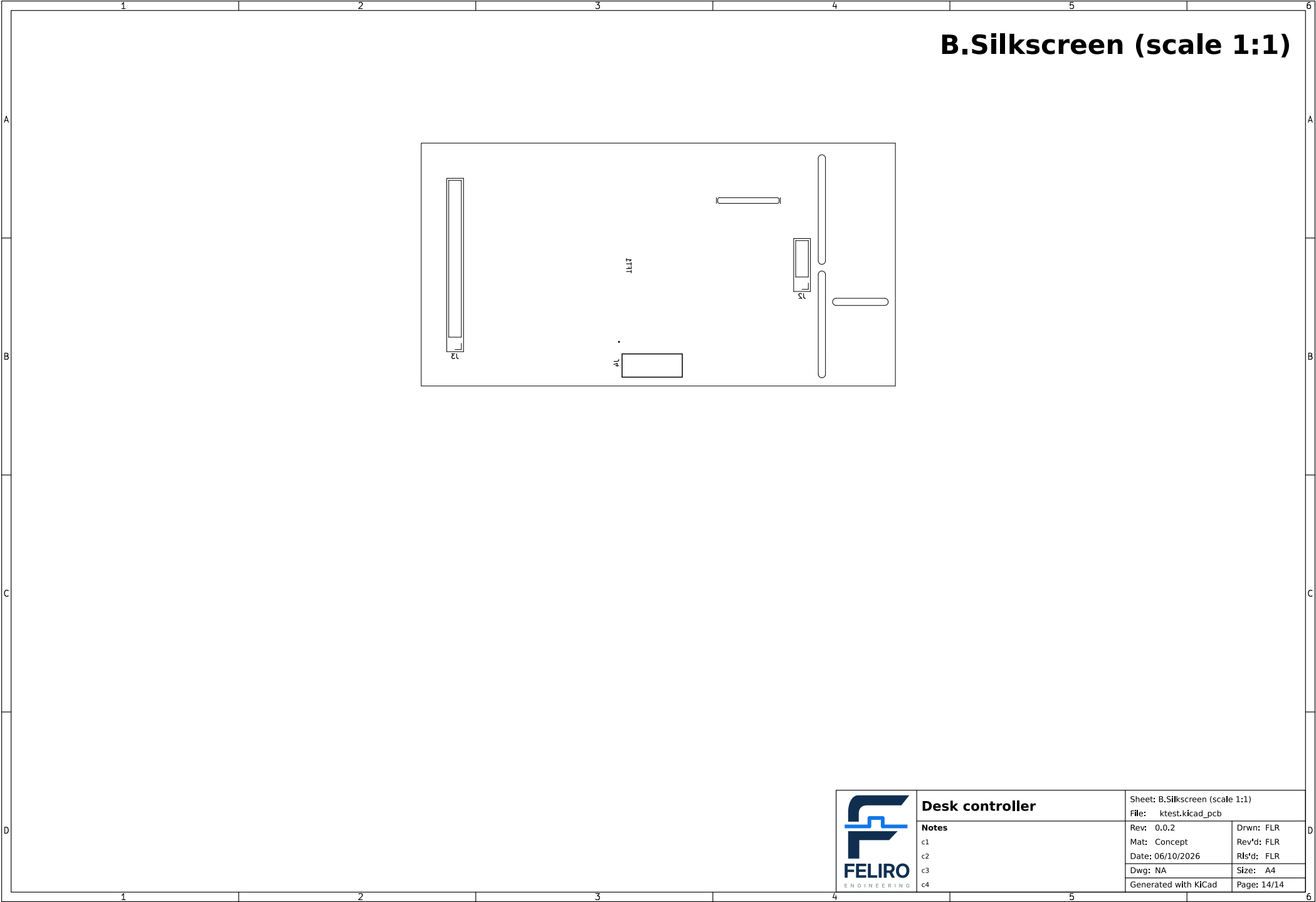
Churras Electronics

Limit sw

Table controller

J6





Desk controller		Sheet: B.Silkscreen (scale 1:1)	
Notes c1 c2 c3 c4		File: ktest.kicad_pcb	
		Rev: 0.0.2	Drwn: FLR
		Mat: Concept	Rev'd: FLR
		Date: 06/10/2026	RLs'd: FLR
		Dwg: NA	Size: A4
		Generated with KiCad	Page: 14/14